

UNG FACILITIES PLANNING & GIS

Version 3.6

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Parses freeform Active Directory location text into structured GIS-ready fields

AD_Parsing_V3.6.py

Purpose

Active Directory stores employee office locations as freeform text in the physicalDeliveryOfficeName field — entered by staff in whatever format they choose. This script is the entry point to the entire pipeline. It reads the raw AD export and parses each location string into three structured fields: Building, Room, and City. Without this step nothing downstream can match an employee to a GIS polygon.

When to Run

- Weekly, as Step 1 of the pipeline, immediately after downloading a fresh AD export from IT
- Any time a new AD export is obtained

Requirements

- Python with pandas installed (no ArcGIS required)
- Raw AD export Excel file from IT (FacStaffWithAddress_MMDDYYYY.xlsx)

Inputs

Item	Value	Notes
Raw AD Export	FacStaffWithAddress_MMDDYY YY.xlsx	Downloaded from IT; contains Value.physicaldeliveryofficename and Value.postaladdress columns
Script config	input_file path at bottom of script	Update filename for each run

What It Does

- Reads every row from the AD export
- Parses Value.physicaldeliveryofficename into Building, Room, and City fields using comma-delimited logic with multiple fallback strategies
- Handles typos in city names (DAHLONGEA, Gainesvllle, etc.) via corrections dictionary
- Handles entries where room and city are space-delimited rather than comma-delimited (e.g. "202 DAHLONGEA")
- Flags Suspect records — part-time, adjunct, eCore, trades, police — who are not expected to have fixed offices
- Parses secondary office from Value.postaladdress when present, appending a second row tagged Source = postaladdress
- Writes a Surname field from Value.sn for downstream matching

Outputs

Item	Value	Notes
Parsed AD Excel	FacStaffWithAddress_MMDDYY YY_parsed.xlsx	Primary output; input to all downstream pipeline steps
Console summary	Printed to screen	Total records, complete records, suspect count, online/blank count

Key Configuration

input_file: Path to raw AD Excel — update the filename for each weekly run

primary_col: Value.physicaldeliveryofficename — do not change

secondary_col: Value.postaladdress — do not change

Suspect Record Logic

Records matching SUSPECT_KEYWORDS (affiliates, part-time, eCore) or NO_OFFICE_TRADES (custodial, grounds, HVAC, police, electrician, plumber) are flagged Suspect = Y only when their location data is also incomplete. If someone in a suspect category has a complete Building + Room + City, they are matched normally. Suspect = Y means suspect title AND incomplete location.

Version Notes

V3.6: Removed DEHYPHEN_BUILDINGS logic that was introduced in V3.5 — the parser does not have access to GIS building codes, so hyphen normalization was moved to the updater scripts where it belongs. DAHLONGEA city correction and space-delimited fallback retained.

V3.5: Added DAHLONGEA city correction and space-delimited city fallback. DEHYPHEN_BUILDINGS introduced but ineffective (see V3.6).